

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf\_736be611-fb5\agent-a2a54f057c25c5767.jsonl`
- SHA-256: `837e3f3a873842c411980c9fe2924e14479390c324614b561b465f4782cc5c36`
- Source modified: `2026-06-16T15:15:57+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf\_736be611-fb5`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T15:14:30.289Z)

Adversarially VERIFY this finding from the real-footage rally-detection validation. Try hard to REFUTE or show it is overstated, using ONLY the data (you MAY Read output/local_vs_gemini_6.json). Default to holds=false if the numbers do not clearly support it.

CLAIM (windowing-TAL): The over-merge -> over-segment pendulum is real and global: the cap=12 fix swung aggregate count from 35% of Gemini to 152% of Gemini, never landing near 1.0x. EVIDENCE CITED: Aggregate count vs Gemini 165: baseline 48 (0.29x, mean window 62.9s, 27 merge-groups, mIoU 0.23) -> W1 250 (1.52x, mean 10.8s, mIoU 0.39) -> W1+W2 222 (1.35x). Baseline collapses entire games into mega-windows (GX010129 = 2 windows, mean 267s, 27 gem rallies in one merge, mIoU 0.0); W1 over-splits. No single global setting hit ~1.0x in aggregate.

REAL-FOOTAGE VALIDATION DATA (6 Kushagra/Yash June-12 clips, 1080p proxies, fully-LOCAL no-AI WASB windowing vs Gemini. Gemini = strong REFERENCE, NOT ground truth ? these 6 have no golden labels EXCEPT mahadevpura-2 which the owner hand-labeled, see GROUND-TRUTH below).

Rally COUNT (Gemini | local baseline | local W1 cap=12 | local W1+W2 mc=1):

GX010128:	39		15		78		75
GX010129:	27		2		50		47
GX020125:	11		6		18		15
mahadevpura-0:	40		19		62		43
mahadevpura-1:	9		1		2		2
mahadevpura-2:	39		5		40		40

AGGREGATE: Gemini 165 | baseline 48 (mean window 62.9s, 27 merge-groups, mIoU 0.23) | W1 250 (mean 10.8s, 10 merge-groups, mIoU 0.39) | W1+W2 222 (mean 11.5s, 10 merge-groups, mIoU 0.39, 6 gemini-only "misses")

Per-video mean window (W1): GX010128 7.8s, GX010129 9.3s, GX020125 14.1s, mahadevpura-0 10.4s, mahadevpura-1 86.9s (the continuously-active blob W1 cannot cap ? no inactivity gap), mahadevpura-2 13.8s. Gemini means ~5.7-7.2s.

GROUND-TRUTH (mahadevpura-2, 40 owner-labeled rallies, mean 5.7s, verified frame-aligned):

F1@0.5 / F1@0.25 / mean-IoU: Gemini 0.557/0.835/0.55 ; local W1+W2 0.150/0.500/0.31 ; local W1+W2+hard-cap 0.296/0.722/0.50 (best local). baseline 0.000/0.000/0.03.

BOUNDARY ERROR (matched rallies, seconds): Gemini |?start| med 1.46 |?end| med 1.19 ; local-hardcap |?start| med 1.74 (~= Gemini!) |?end| med 3.12 (>> Gemini).

=> local's rally-START detection is ~at par with Gemini; the gap is the rally-END.

IoU<->time: for a 5.7s rally, tIoU 0.5 tolerates only ~+-1.9s shift (the ~1-1.5s service window nearly exhausts it -> tIoU 0.5 over-penalizes; owner wants a +-2s tolerance metric).

PRIOR GOLDEN DELTAS (n=6, then n=7 incl. mahadevpura-2): baseline F1 0.300 -> W1 0.439 (+0.139, 6/0)

-> W1+W2 0.457 (+0.019, 6/0); hard-cap golden-neutral on n=6 (+0.006 n.s.) but became BEST on n=7

(continuously-active clips). All flags default-OFF, promoted only on measured LOVO lift + sign-test.

WINDOWING KNOBS: a frame is "active" iff shuttle visible AND per-second-normalized velocity > velocity_thresh(0.02); active frames within merge_gap(2.0s) coalesce; min_window(2.0s). W1 = max_window_duration cap(12s) split at the largest internal inactivity gap. W1.5 hard-cap = midpoint-chop when no gap. W2 = net-crossings gate (drop windows with <1 net crossing).

2. assistant (2026-06-16T15:14:34.411Z)

I'll verify this claim by examining the actual data file.

3. user (2026-06-16T15:14:35.365Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard.

4. assistant (2026-06-16T15:14:37.559Z)

The file isn't at the expected path. Let me locate it.

5. user (2026-06-16T15:14:37.982Z)

No files found

6. user (2026-06-16T15:14:42.005Z)

```
/usr/bin/bash: line 1: cd: C:\Users\avidu\Projects\badminton-highlight-indexer: No such file or directory
---SEARCH---
./eval_baselines/gemini_wrapper_2026-06-10.json
```

7. user (2026-06-16T15:14:47.135Z)

```
backend\eval\gemini_refine.py
backend\pipeline\segmenters\gemini.py
backend\providers\gemini.py
eval_baselines\gemini_wrapper_2026-06-10.json
scratch\gemini_video_debugger.py
tests\test_gemini.py
tests\test_gemini_refine.py
tests\test_provider_gemini.py
tests\__pycache__\test_gemini.cpython-313-pytest-9.0.3.pyc
tests\__pycache__\test_gemini_refine.cpython-313-pytest-9.0.3.pyc
tests\__pycache__\test_provider_gemini.cpython-313-pytest-9.0.3.pyc
backend\eval\__pycache__\gemini_refine.cpython-313.pyc
backend\pipeline\segmenters\__pycache__\gemini.cpython-313.pyc
backend\providers\__pycache__\gemini.cpython-313.pyc
```

8. assistant (2026-06-16T15:14:50.327Z)

The cited data file `output/local\_vs\_gemini\_6.json` does not exist anywhere in the repo or worktree. Let me check the artifacts directory mentioned in git status and look for any related validation output.

9. user (2026-06-16T15:14:52.744Z)

```
total 292
drwxr-xr-x 1 avidu 197609    0 Jun 16 20:34 .
drwxr-xr-x 1 avidu 197609    0 Jun 16 11:24 ..
drwxr-xr-x 1 avidu 197609    0 Jun 16 11:14 chunks
```

```

drwxr-xr-x 1 avidu 197609      0 Jun 16 20:38 highlights_work
-rw-r--r-- 1 avidu 197609 212992 Jun 16 20:34 val.db
-rw-r--r-- 1 avidu 197609 61195 Jun 16 20:38 val_extract.log
drwxr-xr-x 1 avidu 197609      0 Jun 16 20:38 val_reels
drwxr-xr-x 1 avidu 197609      0 Jun 16 20:34 wasb
---OUTPUT DIRS---
C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/agent-a771ff1d85e18f2ed/output
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy/output
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/output
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1/output
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hardchop/output
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/local-noai/output
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier0-eval/output
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing/output
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-netcross/output
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-safeguard/output
C:/Users/avidu/Projects/badminton-highlight-indexer/output
---ARTIFACTS---
C:/Users/avidu/Projects/badminton-highlight-indexer/artifacts/rally_tal_gen0/0.1.0/manifest.json
C:/Users/avidu/Projects/badminton-highlight-indexer/artifacts/rally_tal_gen0/0.1.0/weights.json
C:/Users/avidu/Projects/badminton-highlight-indexer/artifacts/tridet.pdf
C:/Users/avidu/Projects/badminton-highlight-indexer/artifacts/tridet.txt

```

10. assistant (2026-06-16T15:14:55.790Z)

The cited file `output/local\_vs\_gemini\_6.json` is not present in the worktree's `output/` dir. Let me search the other worktrees and the main output dir, since the validation may have been run elsewhere.

11. user (2026-06-16T15:14:58.188Z)

```

C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_vs_gemini_6.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_vs_gemini_6.md
C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_vs_gemini_report.md
C:/Users/avidu/Projects/badminton-highlight-indexer/scratch/compare_local_vs_gemini.py
---ALL JSON in output dirs (excluding wasb internals)---
C:/Users/avidu/Projects/badminton-highlight-indexer/output/ab_experiment_n6.json
C:/Users/avidu/Projects/badminton-highlight-
indexer/output/adarsh_avi_singles_golden_features.json
C:/Users/avidu/Projects/badminton-highlight-
indexer/output/gateA_served_verify/served_gateA_summary.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/gbaaddy_golden_features.json
C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010125_run/GX010125_full_summary.json
C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010125_run/GX010125_rescore_summary.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/human_lovo_manifest.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/kushagra_1280.report.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/kushagra_singles_golden_features.json
C:/Users/avidu/Projects/badminton-highlight-
indexer/output/largetest_doubles_golden_features.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_calib_manifest.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_rally_model.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_vs_gemini_6.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/mahadevpura_2_golden_features.json
C:/Users/avidu/Projects/badminton-highlight-
indexer/output/mahadevpura_singles_golden_features.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/mahadevpura_smoke_180s.report.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/phase3_probe.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/test_large_short_golden_features.json

```

12. assistant (2026-06-16T15:15:01.285Z)

The file exists in the main repo output dir, not the worktree. Let me read it.

13. user (2026-06-16T15:15:01.874Z)

```
1      [
2      {
3          "name": "GX010128",
4          "vid": "db84f3e65318",
5          "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\.claude\\worktrees\\w2-safeguard\\output\\wasb\\GX010128__db84f3e65318_wasb.csv",
6          "n_gem": 39,
7          "play_gem": 228.0,
8          "mean_gem": 5.846153846153846,
9          "variants": {
10             "baseline": {
11                 "n": 15,
12                 "play": 711.4614614614616,
13                 "mean_dur": 47.430764097430774,
14                 "max_dur": 133.2499165832499,
15                 "count_ratio": 0.38461538461538464,
16                 "matched": 4,
17                 "local_only": 1,
18                 "gemini_only": 0,
19                 "splits": 0,
20                 "merges": 10,
21                 "mean_iou": 0.3564105364413525,
22                 "gem_rallies_in_merges": 35
23             },
24             "W1": {
25                 "n": 78,
26                 "play": 605.1051051051056,
27                 "mean_dur": 7.757757757757765,
28                 "max_dur": 16.616616616616625,
29                 "count_ratio": 2.0,
30                 "matched": 35,
31                 "local_only": 34,
32                 "gemini_only": 0,
33                 "splits": 4,
34                 "merges": 0,
35                 "mean_iou": 0.5294639963484731,
36                 "gem_rallies_in_merges": 0
37             },
38             "W1+W2": {
39                 "n": 75,
40                 "play": 594.494494494495,
41                 "mean_dur": 7.926593259926601,
42                 "max_dur": 16.616616616616625,
43                 "count_ratio": 1.9230769230769231,
44                 "matched": 35,
45                 "local_only": 31,
46                 "gemini_only": 0,
47                 "splits": 4,
48                 "merges": 0,
49                 "mean_iou": 0.5294639963484731,
50                 "gem_rallies_in_merges": 0
51             }
52         },
53         "status": "ok"
54     },
55     {
56         "name": "GX010129",
57         "vid": "99a3e73afde3",
58         "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\.claude\\worktrees\\w2-safeguard\\output\\wasb\\GX010129__99a3e73afde3_wasb.csv",
59         "n_gem": 27,
```

```

60     "play_gem": 195.6,
61     "mean_gem": 7.2444444444444445,
62     "variants": {
63         "baseline": {
64             "n": 2,
65             "play": 534.4177510844178,
66             "mean_dur": 267.2088755422089,
67             "max_dur": 522.6893560226894,
68             "count_ratio": 0.07407407407407407,
69             "matched": 0,
70             "local_only": 1,
71             "gemini_only": 0,
72             "splits": 0,
73             "merges": 1,
74             "mean_iou": 0.0,
75             "gem_rallies_in_merges": 27
76         },
77         "w1": {
78             "n": 50,
79             "play": 463.7137137137137,
80             "mean_dur": 9.274274274274273,
81             "max_dur": 27.84451117784451,
82             "count_ratio": 1.8518518518518519,
83             "matched": 23,
84             "local_only": 18,
85             "gemini_only": 0,
86             "splits": 4,
87             "merges": 0,
88             "mean_iou": 0.4819655005521238,
89             "gem_rallies_in_merges": 0
90         },
91         "w1+w2": {
92             "n": 47,
93             "play": 447.0303636970303,
94             "mean_dur": 9.511284333979368,
95             "max_dur": 27.84451117784451,
96             "count_ratio": 1.7407407407407407,
97             "matched": 22,
98             "local_only": 16,
99             "gemini_only": 1,
100            "splits": 4,
101            "merges": 0,
102            "mean_iou": 0.48514996335815863,
103            "gem_rallies_in_merges": 0
104        }
105    },
106    "status": "ok"
107 },
108 {
109     "name": "GX020125",
110     "vid": "407fa7e7e9f1",
111     "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\.claude\\worktrees\\local-noai\\output\\wasb\\GX020125__407fa7e7e9f1_wasb.csv",
112     "n_gem": 11,
113     "play_gem": 71.0,
114     "mean_gem": 6.454545454545454,
115     "variants": {
116         "baseline": {
117             "n": 6,
118             "play": 280.6806806806807,
119             "mean_dur": 46.78011344678012,
120             "max_dur": 82.66599933266599,
121             "count_ratio": 0.5454545454545454,
122             "matched": 1,
123             "local_only": 2,

```

```

124         "gemini_only": 0,
125         "splits": 0,
126         "merges": 3,
127         "mean_iou": 0.40270862800565765,
128         "gem_rallies_in_merges": 10
129     },
130     "w1": {
131         "n": 18,
132         "play": 254.03737070403744,
133         "mean_dur": 14.113187261335414,
134         "max_dur": 40.05672339005673,
135         "count_ratio": 1.6363636363636365,
136         "matched": 9,
137         "local_only": 8,
138         "gemini_only": 0,
139         "splits": 0,
140         "merges": 1,
141         "mean_iou": 0.4329964346464835,
142         "gem_rallies_in_merges": 2
143     },
144     "w1+w2": {
145         "n": 15,
146         "play": 246.5131798465132,
147         "mean_dur": 16.434211989767547,
148         "max_dur": 40.05672339005673,
149         "count_ratio": 1.3636363636363635,
150         "matched": 9,
151         "local_only": 5,
152         "gemini_only": 0,
153         "splits": 0,
154         "merges": 1,
155         "mean_iou": 0.4329964346464835,
156         "gem_rallies_in_merges": 2
157     }
158 },
159 "status": "ok"
160 },
161 {
162     "name": "mahadevpura-0",
163     "vid": "6460c2fc7647",
164     "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\.claude\\worktrees\\w2-safeguard\\output\\wasb\\mahadevpura-0__6460c2fc7647_wasb.csv",
165     "n_gem": 40,
166     "play_gem": 262.0,
167     "mean_gem": 6.55,
168     "variants": {
169         "baseline": {
170             "n": 19,
171             "play": 711.5615615615616,
172             "mean_dur": 37.45060850324008,
173             "max_dur": 134.13413413413414,
174             "count_ratio": 0.475,
175             "matched": 6,
176             "local_only": 4,
177             "gemini_only": 0,
178             "splits": 0,
179             "merges": 9,
180             "mean_iou": 0.3446682872977475,
181             "gem_rallies_in_merges": 34
182         },
183         "w1": {
184             "n": 62,
185             "play": 646.1127794461132,
186             "mean_dur": 10.421173862034085,
187             "max_dur": 47.66433099766434,

```

```

188         "count_ratio": 1.55,
189         "matched": 34,
190         "local_only": 22,
191         "gemini_only": 0,
192         "splits": 2,
193         "merges": 2,
194         "mean_iou": 0.4928109196809172,
195         "gem_rallies_in_merges": 4
196     },
197     "W1+W2": {
198         "n": 43,
199         "play": 549.5161828495164,
200         "mean_dur": 12.779446112779452,
201         "max_dur": 47.66433099766434,
202         "count_ratio": 1.075,
203         "matched": 30,
204         "local_only": 9,
205         "gemini_only": 5,
206         "splits": 1,
207         "merges": 2,
208         "mean_iou": 0.4837616679798798,
209         "gem_rallies_in_merges": 4
210     }
211 },
212 "status": "ok"
213 },
214 {
215     "name": "mahadevpura-1",
216     "vid": "38bef75d78cb",
217     "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\.claude\\worktrees\\local-noai\\output\\wasb\\mahadevpura-1_38bef75d78cb_wasb.csv",
218     "n_gem": 9,
219     "play_gem": 51.2,
220     "mean_gem": 5.688888888888889,
221     "variants": {
222         "baseline": {
223             "n": 1,
224             "play": 174.2742742742743,
225             "mean_dur": 174.2742742742743,
226             "max_dur": 174.2742742742743,
227             "count_ratio": 0.1111111111111111,
228             "matched": 0,
229             "local_only": 0,
230             "gemini_only": 0,
231             "splits": 0,
232             "merges": 1,
233             "mean_iou": 0.0,
234             "gem_rallies_in_merges": 9
235         },
236         "W1": {
237             "n": 2,
238             "play": 173.79045712379047,
239             "mean_dur": 86.89522856189524,
240             "max_dur": 101.88521855188523,
241             "count_ratio": 0.2222222222222222,
242             "matched": 0,
243             "local_only": 0,
244             "gemini_only": 0,
245             "splits": 0,
246             "merges": 2,
247             "mean_iou": 0.0,
248             "gem_rallies_in_merges": 9
249         },
250         "W1+W2": {
251             "n": 2,

```

```

252         "play": 173.79045712379047,
253         "mean_dur": 86.89522856189524,
254         "max_dur": 101.88521855188523,
255         "count_ratio": 0.2222222222222222,
256         "matched": 0,
257         "local_only": 0,
258         "gemini_only": 0,
259         "splits": 0,
260         "merges": 2,
261         "mean_iou": 0.0,
262         "gem_rallies_in_merges": 9
263     }
264 },
265     "status": "ok"
266 },
267 {
268     "name": "mahadevpura-2",
269     "vid": "038e0e0915a2",
270     "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\.claude\\worktrees\\local-noai\\output\\wasb\\mahadevpura-2__038e0e0915a2_wasb.csv",
271     "n_gem": 39,
272     "play_gem": 266.0,
273     "mean_gem": 6.82051282051282,
274     "variants": {
275         "baseline": {
276             "n": 5,
277             "play": 605.9059059059059,
278             "mean_dur": 121.18118118118119,
279             "max_dur": 297.03036369703034,
280             "count_ratio": 0.1282051282051282,
281             "matched": 2,
282             "local_only": 0,
283             "gemini_only": 0,
284             "splits": 0,
285             "merges": 3,
286             "mean_iou": 0.2945165777074605,
287             "gem_rallies_in_merges": 37
288         },
289         "W1": {
290             "n": 40,
291             "play": 551.0844177510842,
292             "mean_dur": 13.777110443777104,
293             "max_dur": 43.3266599933267,
294             "count_ratio": 1.0256410256410255,
295             "matched": 12,
296             "local_only": 4,
297             "gemini_only": 0,
298             "splits": 4,
299             "merges": 5,
300             "mean_iou": 0.4204774017180335,
301             "gem_rallies_in_merges": 11
302         },
303         "W1+W2": {
304             "n": 40,
305             "play": 551.0844177510842,
306             "mean_dur": 13.777110443777104,
307             "max_dur": 43.3266599933267,
308             "count_ratio": 1.0256410256410255,
309             "matched": 12,
310             "local_only": 4,
311             "gemini_only": 0,
312             "splits": 4,
313             "merges": 5,
314             "mean_iou": 0.4204774017180335,
315             "gem_rallies_in_merges": 11

```



```
316         }
317     },
318     "status": "ok"
319 }
320 ]
```

14. assistant (2026-06-16T15:15:04.912Z)

Now I have the raw data. Let me verify each numeric claim carefully. Let me compute the aggregates directly from the JSON.

15. user (2026-06-16T15:15:12.632Z)

GEM total: 165
BASELINE n: 48 ratio 0.2909 merges 27 agg mean window 62.88
W1 n: 250 ratio 1.5152 merges 10 agg mean window 10.78
W1+W2 n: 222 ratio 1.3455 merges 10 gem_only 6 agg mean window 11.54

Per-video W1 mean_dur:

GX010128	7.8
GX010129	9.3
GX020125	14.1
mahadevpura-0	10.4
mahadevpura-1	86.9
mahadevpura-2	13.8

GX010129 baseline: {'n': 2, 'play': 534.4177510844178, 'mean_dur': 267.2088755422089, 'max_dur': 522.6893560226894, 'count_ratio': 0.07407407407407407, 'matched': 0, 'local_only': 1, 'gemini_only': 0, 'splits': 0, 'merges': 1, 'mean_iou': 0.0, 'gem_rallies_in_merges': 27}

16. assistant (2026-06-16T15:15:16.241Z)

Every cited number checks out against the raw data. Let me now scrutinize the claim's mIoU figures and the per-video count breakdown, since the claim cites specific mIoU aggregates (baseline 0.23, W1 0.39, W1+W2 0.39) that I haven't yet verified.

17. user (2026-06-16T15:15:22.060Z)

baseline unweighted mIoU 0.233 matched-weighted 0.345
W1 unweighted mIoU 0.393 matched-weighted 0.49
W1+W2 unweighted mIoU 0.392 matched-weighted 0.488

name	gem	base	W1	W1+W2
GX010128	39	15	78	75
GX010129	27	2	50	47
GX020125	11	6	18	15
mahadevpura-0	40	19	62	43
mahadevpura-1	9	1	2	2
mahadevpura-2	39	5	40	40

GX010129 baseline: n= 2 mean_dur= 267.2 gem_in_merge= 27 mIoU= 0.0

18. assistant (2026-06-16T15:15:26.087Z)

All numbers verified. Let me now adversarially probe the strongest counter-argument: the claim says "never landing near 1.0x" globally, but the data shows individual videos DO land near 1.0x under W1+W2 (mahadevpura-0 = 1.075x, mahadevpura-2 = 1.026x). I need to check whether the claim's scope ("in aggregate" / "no single global setting") is honest about this.

19. user (2026-06-16T15:15:29.055Z)

Per-video count_ratio (vs Gemini):

name	base	W1	W1+W2
GX010128	0.385	2.0	1.923

GX010129 0.074 1.852 1.741
GX020125 0.545 1.636 1.364
mahadevpura-0 0.475 1.55 1.075
mahadevpura-1 0.111 0.222 0.222
mahadevpura-2 0.128 1.026 1.026

baseline aggregate ratio 0.2909
W1 aggregate ratio 1.5152
W1+W2 aggregate ratio 1.3455

20. assistant (2026-06-16T15:15:41.076Z)

I have everything needed to render a verdict. Let me assess the claim precisely against the data.

The CLAIM states the pendulum is "real and global" and that cap=12 swung aggregate count "from 35% of Gemini to 152% of Gemini, never landing near 1.0x." Let me check each load-bearing element:

****Verified true:****

- Aggregate baseline = 48 of 165 = 0.29x. The EVIDENCE-CITED text says "0.29x" ? correct. But the CLAIM headline says "35% of Gemini." 48/165 = 29.1%, not 35%. This is a discrepancy within the claim itself.
- W1 aggregate = 250 = 1.5152x ? 152%. Correct.
- W1+W2 = 222 = 1.3455x ? 1.35x. Correct.
- Baseline: mean window 62.9s (62.88 ?), 27 merge-groups ?, mIoU 0.23 (unweighted 0.233 ?).
- W1: mean 10.8s (10.78 ?), mIoU 0.39 ?.
- GX010129 baseline = 2 windows, mean 267s ?, 27 gem rallies in one merge ?, mIoU 0.0 ?.
- Per-video W1 mean windows all match; mahadevpura-1 = 86.9s continuously-active blob ?.
- "No single global setting hit ~1.0x in aggregate": baseline 0.29x, W1 1.52x, W1+W2 1.35x ? none near 1.0x in aggregate. ?

****The "35%" discrepancy:**** The CLAIM headline says baseline was "35% of Gemini" but the data and the claim's own EVIDENCE say 29% (0.29x). 35% is wrong; the correct figure is ~29%.

****The "global" qualifier ? adversarial check:**** Per-video, two clips DO land near 1.0x under W1+W2 (mahadevpura-0 = 1.075x, mahadevpura-2 = 1.026x). So "never landing near 1.0x" is true only in aggregate, not per-video. The claim is careful to scope this with "in aggregate" / "no single global setting hit ~1.0x in aggregate," so the core thesis survives ? but "global" is slightly generous given per-video near-misses exist. The aggregate pendulum (0.29x to 1.52x, skipping ~1.0x) is genuinely real and well-supported.

The thesis holds; one cited number (35%) is wrong and contradicts the claim's own evidence block (29%).

21. user (2026-06-16T15:15:50.024Z)

Structured output provided successfully

22. assistant (2026-06-16T15:15:57.504Z)

I verified the claim against the actual data in `C:\Users\avidu\Projects\badminton-highlight-indexer\output\local\_vs\_gemini\_6.json` (note: the cited path `output/local\_vs\_gemini\_6.json` resolves to the main repo, not the w2-safeguard worktree ? the worktree's `output/` has no such file).

Verdict: holds = true, with one corrected number.

The pendulum thesis is genuinely supported: aggregate count vs Gemini's 165 swings baseline 48 (0.29x, mean window 62.88s, 27 merge-groups, mIoU 0.233) to W1 250 (1.52x, mean 10.78s, mIoU 0.393) to W1+W2 222 (1.35x), never landing near 1.0x in aggregate. The mega-window collapse (GX010129 = 2 windows, mean 267s, 27 Gemini rallies in one merge, mIoU 0.0) and all per-video W1 mean windows match the data exactly.

Two adversarial findings:

1. The headline "35% of Gemini" is wrong ? $48/165 = 29.1\%$, and the claim's own EVIDENCE block correctly says 0.29x. Corrected to 29%.
2. "Global" is slightly overstated: the skip-past-1.0x is strictly aggregate. Per-video, mahadevpura-0 (1.075x) and mahadevpura-2 (1.026x) do land near 1.0x under W1+W2, so no single global setting hits 1.0x only in the aggregate sense ? which the claim does scope correctly.